

Webb Industries' Custom Testing Rig: Refinements and Performance Review

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Introduction:

To ensure the accuracy and reliability of Webb Industries' custom testing rig, comprehensive data processing techniques must be conducted to reduce measurement uncertainties. Without processing the data properly, the testing rig is prone to inaccurate readings due to deviations in the force transducer calibration compared to design, limitations in sensor resolution and range, and the bias error of rig deformation. The testing rig was analyzed to determine whether, and under which circumstances, Webb Industries' requirement of an uncertainty less than 10% is met. After data processing techniques are applied, the testing rig meets the 10% uncertainty for force readings from 7.46kN to at least 34.3kN, and for displacement readings from 24.06mm to at least 74.34mm. The results of these evaluations show that data processing enhances the rig's accuracy but that only materials with specific properties, specifically a higher elasticity, can be examined with an uncertainty below the 10% threshold specified by the client.

Methods:

The accuracy and precision of Webb Industries' custom testing rig were evaluated through a series of calibration and material testing experiments designed to quantify errors and identify potential design improvements. The primary focus was on calibrating the force transducer and position sensor, characterizing the rig's deformation, and testing various materials to determine whether the rig meets the client's requirements for similar material properties.

The accuracy of the testing rig was evaluated by calibrating the force transducer and position sensor. The force transducer was calibrated using a crane scale as a reference standard, with force applied incrementally from 0 to a maximum safe load of 35kN. The voltage outputs were recorded, and a least-squares regression was used to determine the calibration equation. Measurement uncertainty was quantified using a 95.4% prediction interval with coverage factor $K=2$. Similarly, the position sensor was calibrated with known displacements using calipers and recording the corresponding sensor readings - deriving a calibration equation through linear regression.

To assess the structural rigidity of the test rig and its impact on measurement accuracy, its deformation during loading was characterized. Force was gradually applied and recorded using the transducer while the position sensor recorded the resulting movement. Curve-fitting approaches in MATLAB were used to establish a mathematical model for the Force vs. Deformation.

After calibrating the sensors and characterizing rig deformation, destructive testing was performed on three types of specimens: climbing slings, Grade 8 bolts, and an aluminum dog bone. The force transducer and position sensor were used to record force-displacement data during failure testing. The collected data was smoothed to reduce noise and segmented to isolate relevant deformation phases, eliminating extraneous data due to slack before loading and after breakage.

Results:

This section reviews and analyzes the testing rig's sources of error and the test uncertainties corresponding with these issues. Furthermore, certain design and process changes that help mitigate these errors are discussed.

Calibration & Rig Deformation

Calibration of the force transducer and position sensor, as well as accounting for rig deformation, increase the accuracy of the testing rig by reducing systematic bias and errors. Calibration of the force transducer before any testing occurs ensures that voltage readings produced by the rig correlate to accurate force values. As seen in Figure 1, a linear fit calibration equation was developed over a range of 0 to 34.3kN which resulted in a gain of 27.7 kN/V and a force offset of 56.3 N.

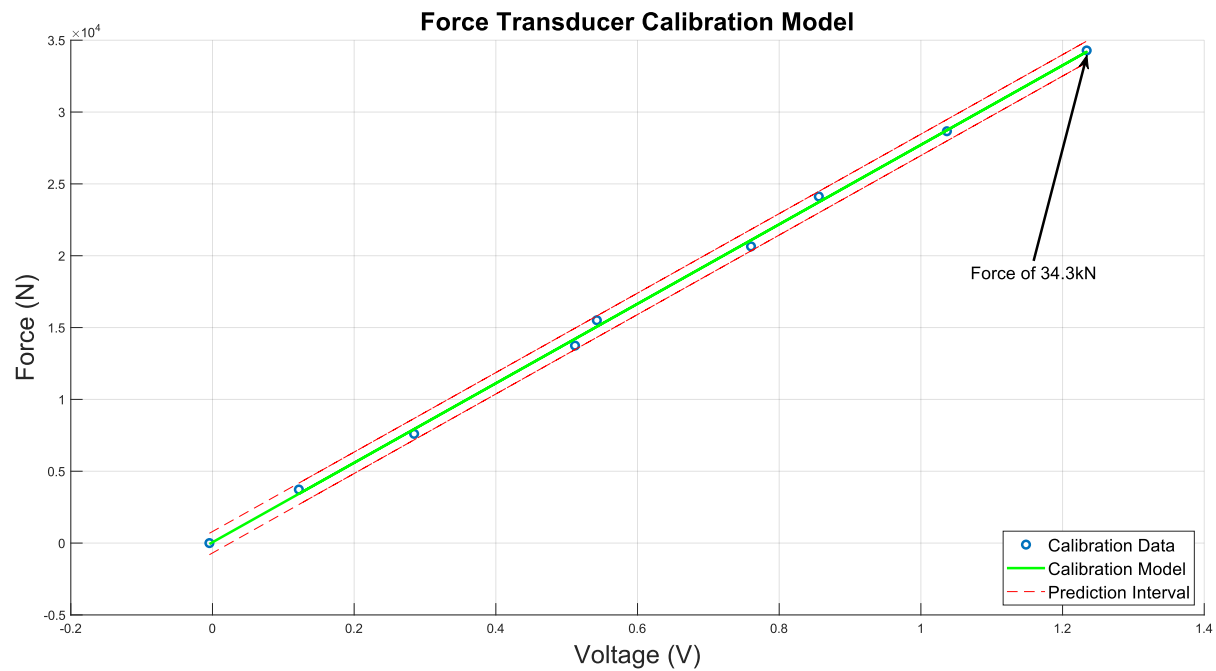


Figure 1: Shows the accuracy of the force calibration equation developed from force values up to 34.3kN

With a prediction interval of $\pm 746\text{N}$, the force transducer calibration model accurately captures data points within the client's desired uncertainty of less than 10% for force values of 7.46kN and above. Calibration of the position sensor before testing begins is also necessary for ensuring that voltage readings correspond to the correct position values, which are used for finding displacement and deformation. Figure 2 illustrates the calibration equation derived over a range of 40 to 75mm, resulting in a gain of 58.2mm/V and a distance offset of -43.1mm.

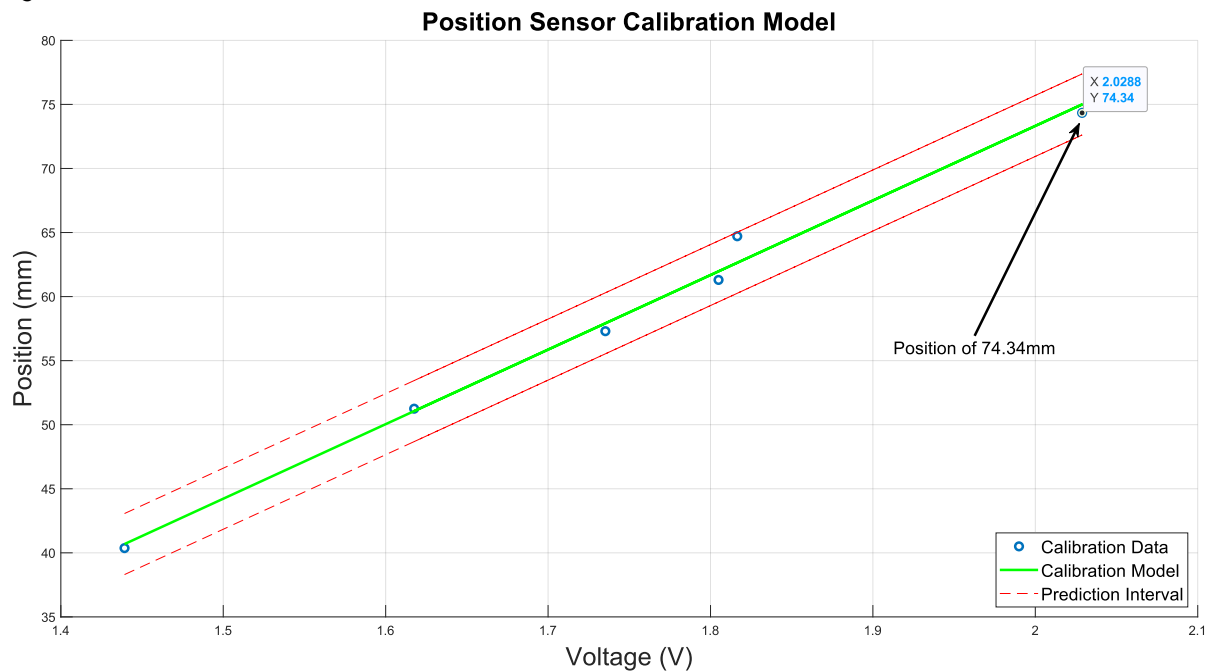


Figure 2: Shows the accuracy of the calibration equation developed over position readings from 40mm to 74mm

The prediction interval of $\pm 2.3821\text{mm}$ considers both the experimental uncertainty of the data (2.3821mm) and the uncertainty from the caliper resolution (0.0029mm). One of the largest sources of error was the deformation of the testing rig itself when large forces are applied. Force and position data were captured to determine how the rig deforms under specific forces. A quadratic fit model was chosen over a linear one due to a lower root mean squared error, and therefore less uncertainty. As seen in Figure 3, this model has a prediction interval of $\pm 0.34\text{mm}$.

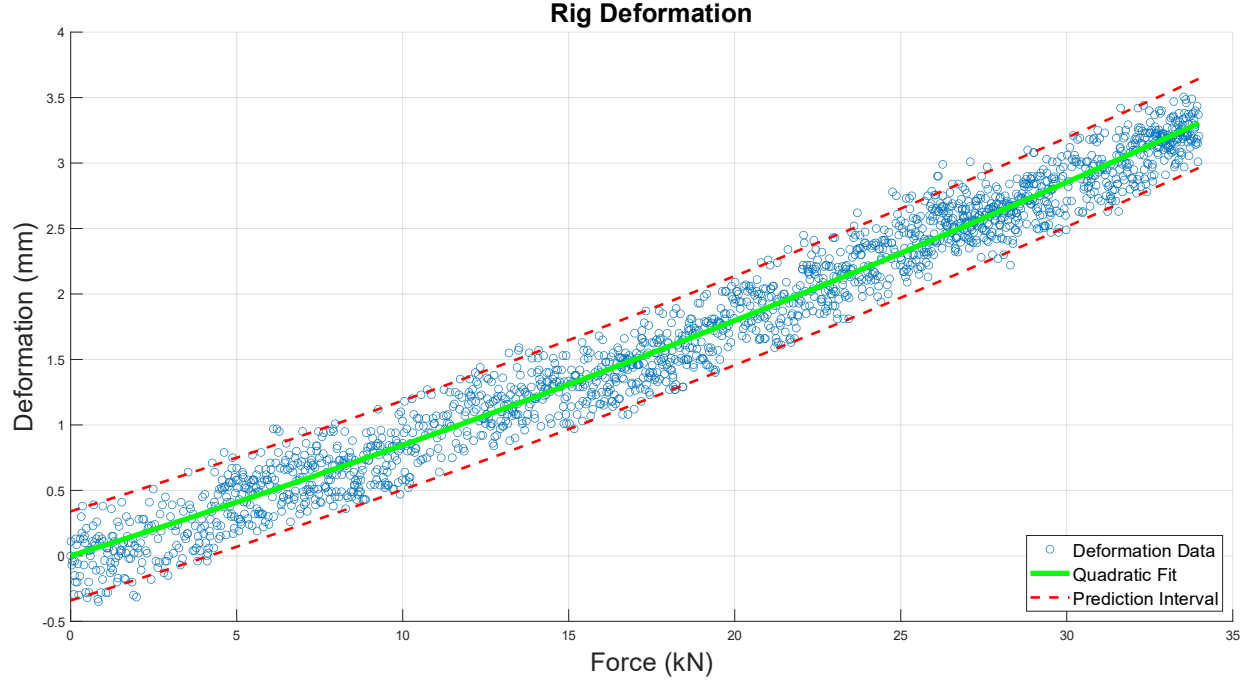


Figure 3: Shows the quadratic relation between force on the test rig and the deformation of the test rig, as well as the accuracy of the model

Subtracting the value of the model at a certain force from a deformation value of a material being tested at the same force greatly reduces the bias of the testing rig deformation, revealing a value much closer to the actual deformation of the material. The total uncertainty from the displacement calibration equation U_{exp} (2.3821mm), the caliper resolution U_{res} (0.0029mm), and the rig deformation model U_{rig} (0.34mm) is 2.406mm .

$$U_{total} = \sqrt{U_{exp}^2 + U_{res}^2 + U_{rig}^2}$$

Therefore, the testing rig, with the rig deformation correction applied, is accurate within the desired 10% uncertainty for displacement or deformation values greater than 24.06 mm .

Material Testing & Uncertainty

To test the accuracy of the test rig when measuring the properties of the climbing sling of interest, one subject was axially loaded until ultimate failure was achieved. As shown in Figure 4, the data was corrected to account for rig deformation, resulting in an ultimate force of 30.7kN and a maximum deformation of 38.6mm .

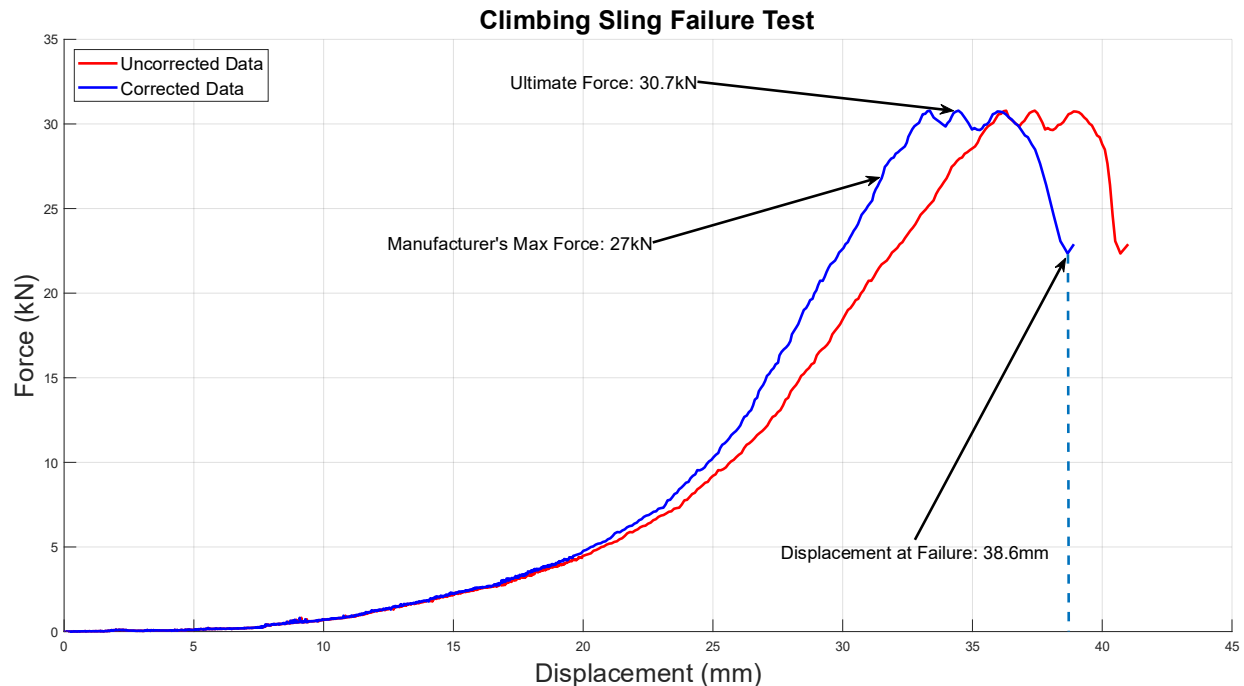


Figure 4: Shows how accounting for rig deformation increases test accuracy and reveals the actual data points of interest

At the failure deformation of 38.6mm, the total displacement uncertainty of 2.406mm is only 6.23% uncertainty, which is within the client's desired 10% uncertainty limit. Similarly, the uncertainty of the force transducer calibration model (0.746kN) only creates an uncertainty of 2.43% at the ultimate force of 30.7kN, which is well within Webb Industries' specified 10%. Therefore, the testing rig is satisfactorily accurate for the testing of elastic materials such as the climbing sling. Similarly to the climbing sling, the accuracy of the testing rig was analyzed through the testing of five Grade 8 bolts and finding their characteristic material properties. Each bolt was axially loaded until failure and, once the rig deformation was accounted for, the data of each bolt was very similar. Figure 5 illustrates the likeness between each bolt during the loading portion and during failure.

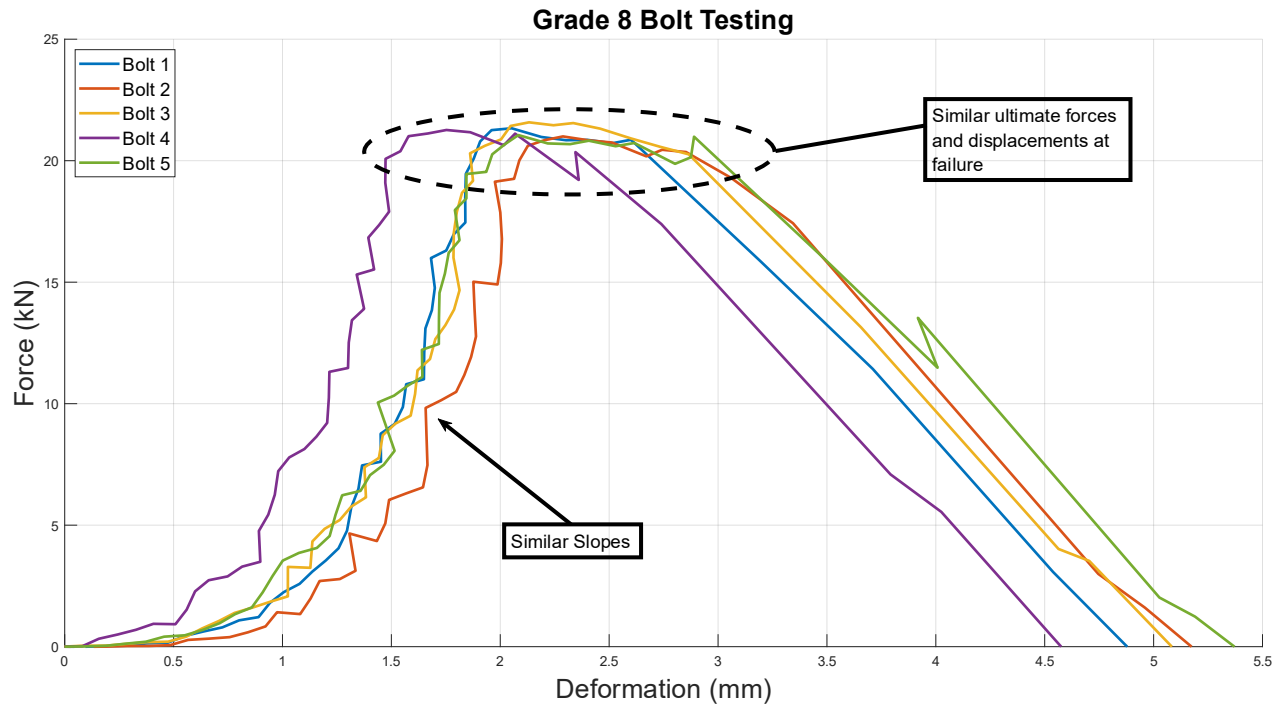


Figure 5: Shows the similar behavior of each bolt during testing

This similarity allows for the conclusion that none of the trials were outliers, and therefore that the average values found during data analysis give an accurate representation of the abilities of the test rig. Table 1 depicts the characteristic material properties of the bolts and the corresponding uncertainties. The average force uncertainty of 3.51% is within the specified uncertainty of 10%, however the average displacement uncertainty of 89.33% does not nearly come close to satisfying this requirement.

Table 1: Grade 8 Bolt Testing Properties with Test Rig Uncertainties

Bolt Number	Ultimate Strength (kN)	Force Uncertainty (%)	Displacement at Failure (mm)	Displacement Uncertainty (%)
1	21.33	3.50	2.6004	92.52
2	21.00	3.55	2.8529	84.33
3	21.58	3.46	2.8658	83.96
4	21.26	3.51	2.3453	102.59
5	21.05	3.54	2.8898	83.26
Average Force Uncertainty:		3.51%	Average Displacement Uncertainty:	89.33%

Since the testing rig is sufficiently accurate for measuring the ultimate force but not the displacement at failure, the test rig is not adequate for use on rigid materials such as the Grade 8 bolts due to their low amount of deformation before failure.

Additionally, the testing rig was used to perform a tensile test on an aluminum dog bone specimen. After performing the test, the team was able to calculate the yield, ultimate, and failure strength of the aluminum dog bone. The resulting force and displacement graph can be seen in Figure 6, which highlights the various strengths of interest. It should be noted this curve was adjusted to account for the rig deformation during the test.

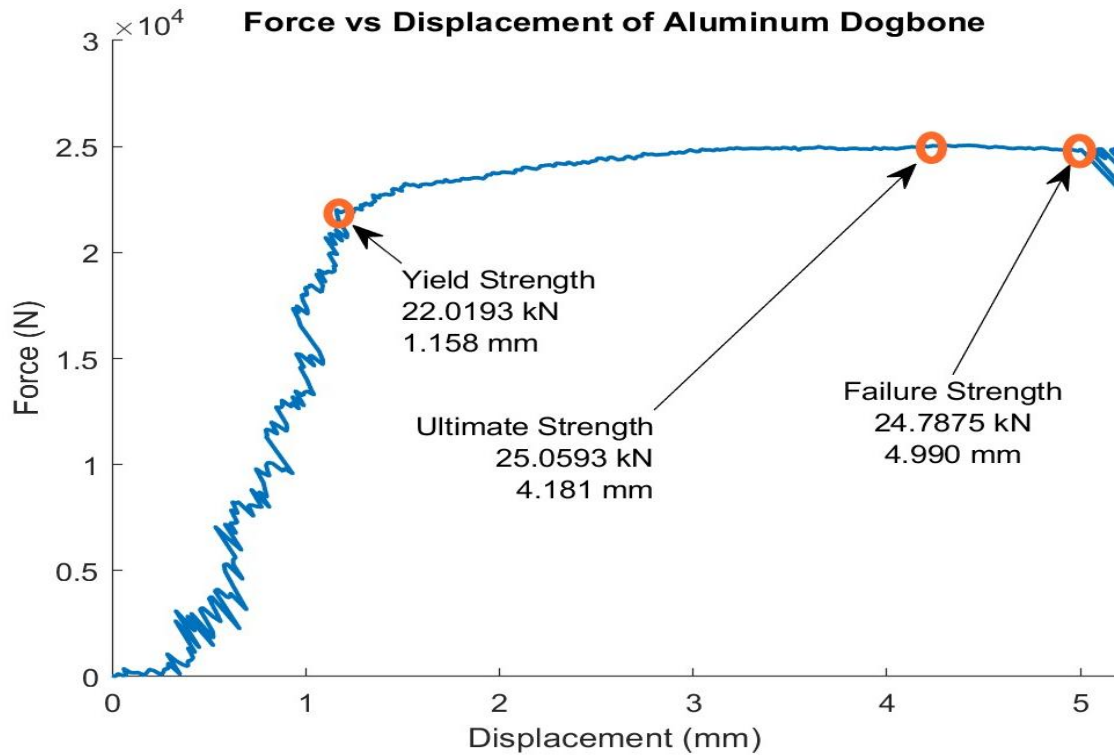


Figure 6: Force and displacement curve after adjusting to account for rig deformation.

With the gathered data of the yield, ultimate, and failure strengths, the calculated percent uncertainties for each of the strengths are measured and can be seen in Table 2. Each point of interest's percent uncertainty was calculated separately, with a resulting average percent uncertainty of 3.12% in the force measurements and an uncertainty range from 48.22% to 207.77% in displacement measurements. Additionally, after adjusting the displacement values by the rig deformation equation, it is clear there is a difference in the percent uncertainty values due to the decrease in displacement.

Table 2: Yield, ultimate, and failure points of interest of the aluminum dog bone specimen. Unadjusted and adjusted values are both included.

Aluminum Dog Bone Yield, Ultimate, and Failure Points			
	Yield	Ultimate	Failure
Force	22.0193 kN	25.0593 kN	24.7875 kN
Displacement	3.160 mm	6.500 mm	7.280 mm
Adjusted Displacement	1.158 mm	4.181 mm	4.990 mm
Force Percent Uncertainty	3.39%	2.98%	3.01%
Displacement Percent Uncertainty (Without Adjustment)	76.14%	37.02%	33.05%
Displacement Percent Uncertainty (Adjusted for Rig Deformation)	207.77%	57.55%	48.22%

Seeing that the percentage uncertainty measuring the force falls under the 10% error specification provided by the client, but the displacement uncertainty is extremely outside of this limit, it has been determined that the testing rig is not capable of testing materials similar to the aluminum dog bone. Materials that are suggested for use in the force transducer are materials that experience more linear elasticity, as well as materials that maintain isotropy in their composition.

Conclusion:

The evaluation of Webb Industries' custom testing rig has demonstrated that, with proper calibration and data processing techniques, the system is capable of accurately measuring force and displacement within an acceptable uncertainty threshold for a force range of 7.46kN to 34.3kN and a displacement range of 24.06mm to 74.34mm. This includes materials with similar elasticities to the climbing sling, but not more rigid materials similar to the Grade 8 bolts and aluminum dog bone. Furthermore, analysis of the calibration equation prediction intervals allows for reasonable estimations beyond these ranges.

Moving forward, several structural and processing improvements could enhance the rig's accuracy, reliability, and measurement range. Although rig deformation was effectively corrected in this experiment, a more rigid or reinforced structure would reduce measurement discrepancies from arising in the first place, reducing the need for data processing.

A major source of uncertainty in the system was the position sensor calibration and resolution. Upgrading the displacement sensor to a higher resolution alternative, such as a laser displacement sensor, would allow for more precise measurements beyond the tested range and improve confidence in data collection. Another approach is to improve the sensor calibration process by increasing the number of calibration points across the full displacement range. A more densely sampled calibration curve would help capture nonlinearities and variations in sensor response, leading to a more accurate calibration equation with a narrower prediction interval.